

Abdoul Hamid COULIBALY

Backend Engineer — Data Systems & Cloud

📍 Rennes, France

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Go and Python backend engineer with 4+ years of experience on distributed data platforms. Real-time ingestion, microservices, event-driven architectures. I've worked on both pipeline design and production operations: on-call duty, incident resolution, observability.

Professional Experience

Backend Engineer – Data Supply Chain

November 2023 - Present

Univers (QoS Energy) | La Chapelle-sur-Erdre, France

Architecture & Data Ingestion

- Development and refactoring of Go and Python microservices for ingesting millions of metrics per minute, in a GitOps-managed Kubernetes environment.
- Implementation and extension of event-driven architectures (RabbitMQ, NATS) for ingestion and asynchronous processing.
- Design of resilient pipelines handling network issues and source instability: distributed error handling, retry, backfilling and manual reload.

Integration & Connectors

- Development of connectors for heterogeneous sources: SCADA, REST, FTP, SQL databases.
- Active participation in architecture decisions: technology evaluation (NATS, Redis), data modeling and source API rate limiting management.
- Rewriting Go services to Python to align with Bazefield stack constraints.

Scalability & Resilience

- Service instrumentation with Prometheus and Grafana dashboard setup for operational monitoring.
- On-call rotation (1 week/6): critical incident resolution and L3 support for client technical operations.

Go Python Ruby RabbitMQ NATS PostgreSQL Kubernetes Azure GitOps Prometheus Grafana

Software Engineer

October 2021 - October 2023

NumoData (formerly EXFO Solutions) | Rennes, France

Backend Development

- Development of Go backend services for a network monitoring platform processing millions of metrics per minute.
- Design of analytical queries (SQL, Druid): aggregations, multidimensional queries, optimization via indexing and caching.

Smart Usage Monitoring — Technical Ownership

- End-to-end design and development of a dashboard usage analytics service: most viewed metrics, highest-volume dimensions, detection of underutilized data.
- Results exposed via REST API, Prometheus instrumentation and Grafana dashboards, used for cache optimization and storage cost reduction.

Performance & Infrastructure

- Load testing and correlation analysis of volume/resources (CPU, RAM, latency) for client sizing.
- Development of an internal tool for automatic Kubernetes (Helm) configuration generation based on client sizing results.

Go Python Kafka Apache Druid AWS Prometheus Grafana Helm Kubernetes

Web Developer – Internship

Resalocal | Talloires-Montmin, France

May 2021 - August 2021

- Development of an ingestion pipeline from the Apidae tourism database to the Akeneo PIM: import of POIs, events and activities.
- Bidirectional synchronization: returning enriched data to Apidae after processing.
- POI media management via Cloudinary API: image upload and transformation.
- Handling ingestion constraints: pagination, rate limiting, and preservation of existing PIM data.

PHP

Laravel

Docker

Kubernetes

Akeneo

Cloudinary

REST API

Technical Skills

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Languages

Go

Python

Ruby

TypeScript

SQL

Bash

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Cloud & DevOps

AWS

Azure

Docker

Kubernetes

Terraform

Helm

GitLab CI

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Data & Messaging

PostgreSQL

MySQL

Apache Druid

Kafka

RabbitMQ

NATS

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Observability

Prometheus

Grafana

OpenTelemetry

Education

Master’s in Computer Science – DABI (Decision-Making, Learning & Big Data) | University of Rennes 1

2021 – 2023

Data warehousing, data mining, Cloud & Big Data, artificial intelligence, application integration

Bachelor’s in Computer Science – IT Management | University of Rennes 1

2020 – 2021

IT/management dual competency, databases, web development, object modeling, data analysis

Higher Technician Diploma in Computer Science | INP-HB

2016 – 2019

Algorithms, networking, web development, system administration, project management

General & Professional English Training | Wall Street English

2024

Oral and written communication, IT technical vocabulary, professional presentations, meetings in English

Languages

French (native)

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English (fluent)

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Bambara (native)